STEPHEN NYARANGA

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stephen-nyaranga.github.io | linkedin.com/in/stephennyaranga | ORCID 0009-0001-2577-6994 | Kenyan

Climate scientist and geophysicist. My doctoral research combines Sentinel-1 SAR and deep learning for flood extent mapping, flood-depth estimation and urban resilience in Rwanda, as part of the CLARITY-Africa programme.

EDUCATION

PhD in Climate Science -- CLARITY-Africa Fellowship *Oct 2025 -- Present*
AIMS-RIC Doctoral Training Program, Kigali | Host university: Jomo Kenyatta University of Agriculture and Technology (JKUAT), Dept. of Geomatic Engineering & GIS | Imperial College London

- Full 36-month CLARITY-Africa scholarship awarded by AIMS-RIC (see Research Affiliations).
- Research: Integrating Deep Learning and SAR Analysis for Flood Extent Mapping, Depth Estimation, and Urban Resilience in Musanze and Kigali, Rwanda.
- Supervisors: Prof. Dr. Mouhamadou Bamba Sylla (AIMS-RIC, main); Dr. Ben Howard and Prof. Dr. Wouter Buytaert (Imperial College London); Dr. Mercy Mwaniki (JKUAT, host).

MSc Geophysics (Pursued) *Jan 2023 -- Dec 2024*
ICTP-East African Institute for Fundamental Research (ICTP-EAIFR), Kigali, Rwanda

- Thesis: Time Series InSAR Analysis over the Olkaria Geothermal Power Plant in Kenya: Insights From a 2020--2024 Sentinel-1 InSAR Survey.
- Coursework: seismology, geodesy, InSAR, computational geodynamics, earthquake mechanics, atmospheric physics, GPS geodesy, physics of volcanoes.

MSc Mathematical Sciences -- Climate Science / Data Science *Oct 2021 -- Jul 2022*
African Institute for Mathematical Sciences (AIMS-Senegal), Mbour, Senegal

- Graduated with Distinction. Mastercard Foundation Scholar.
- Dissertation: Atmospheric Rivers and Surface Melting over the Pine Island Glacier in 1983 and 1998.

BSc Education Science (Physics / Mathematics) *Aug 2016 -- Mar 2021*
Karatina University, Nyeri, Kenya

- First Class Honours. Registered with the Teachers Service Commission (TSC), Kenya.

RESEARCH AFFILIATIONS

CLARITY-Africa PhD Fellow in Climate Science *Oct 2025 -- Present*
Climate Change Resilient Equitable Healthy Cities in Africa (CLARITY-Africa) | Imperial College London and AIMS-RIC, Kigali

- CLARITY-Africa is a Wellcome-funded project led by Imperial College London with African partners, including AIMS, on the health impacts of climate change in Accra, Tamale, Kigali and Musanze.
- Doctoral research within the programme on flood risk in Rwanda: SAR flood mapping, field validation and flood-depth estimation for Kigali and Musanze.
- Attended the CLARITY-Africa Annual General Meeting (Jun 2026) and presented a PhD progress update.
- *This work was made possible by CLARITY-Africa provided through the AIMS Research and Innovation Centre.*

Member, AXIOM Research Group *Current*
AIMS-RIC, Kigali

- Designed and built the group's website (axiomresearchs.org).

RESEARCH EXPERIENCE

Field campaign for flood validation data *Apr -- May 2026*
Rwanda

- Recorded flood marks, set up measurement equipment and installed sensors at river crossings to validate Sentinel-1 flood extents and calibrate flood-depth estimates.

PRESENTATIONS

GeoAI Africa Community Meeting -- invited talk (online) *24 Sep 2026*
Sentinel-1 SAR flood mapping in Rwanda. A lightning talk had been accepted for the GeoAI for a Climate-Resilient Africa workshop at Deep Learning Indaba 2026 and was moved here.

2nd Rwanda Climate Science Symposium (RCSS 2026) -- oral presentation, Kigali *22--23 Sep 2026*
Session B: Climate Data and Monitoring Systems. Certificate of appreciation.

CLARITY-Africa Annual General Meeting -- PhD progress update *10 Jun 2026*
SAR-based flood mapping for Rwandan catchments (Nyabarongo River Basin).

AIMS-RIC Training School: Data Science Methods -- oral presentation *Mar -- Apr 2026*
SAR-based flood mapping and land-cover classification in Rwanda.

TEACHING AND SUPERVISION

Teaching Assistant (upcoming) *13 -- 27 Jan 2027*
East-African School for Young Researchers on Advanced Machine Learning Techniques (EASY-ML), University of Nairobi, Chiromo Campus

Co-supervisor, MSc research project *2026*
AIMS Rwanda

- Student: Richard Usengimana. Spatial Analysis of Child Malnutrition in Rwanda: District-Level Integration of DHS Data and Agro-Climatic Indicators. Primary supervisor: Dr. Lema Seknewna (AIMS-RIC).

Assistant Lecturer (Honorary Academic Title) *Jun 2025 -- Oct 2025*
Vatel Rwanda, Kigali

- Honorary academic title for the 2025--2026 academic year, held alongside my Head of IT role and separate from my employment contract. Delivered ICDL courses to lecturers, staff and students, and supported AI integration in the curriculum.

Physics & Mathematics Teacher *Jan 2021 -- Oct 2021*
Ntanki Secondary School, Meru (Jan -- Apr 2021); Kathiani Boys' High School, Machakos (May -- Oct 2021), Kenya

PROFESSIONAL EXPERIENCE

Head of IT Department (Full-Time) *Jul 2024 -- Oct 2025*
Vatel Rwanda, Kigali

- Led digital transformation of teaching; managed Canvas, Moodle and Triade LMS; built in-house platforms for tracking internships, lecturers, hours and attendance; trained staff and students.

IT Associate -- Mastercard Foundation Associate Enrichment Program *Jul 2023 -- Jul 2024*
Vatel Rwanda, Kigali

- Introduced AI tools in teaching, supported the move to Canvas, and ran training in AI, ICDL and digital literacy.

CONFERENCES, WORKSHOPS & TRAINING

EASY-ML School, University of Nairobi -- teaching assistant (upcoming) *13 -- 27 Jan 2027*

From Satellite Data to Forest Carbon Insights: A Practical GeoAI Bootcamp -- accepted participant (upcoming) *20 -- 22 Oct 2026*
AI.Geolabs, live online.

African Summer School on AI for Weather and Climate Modelling -- accepted participant (upcoming) *19 -- 23 Oct 2026*
ACMAD with Meteo Rwanda and AIMS; AIMS Kigali.

PRESAC-22 Regional Climate Outlook Forum for Central African States -- attended as a doctoral researcher *17 -- 21 Aug 2026*
African Union Commission (ClimSA programme), Kigali.

EarthScope: InSAR Processing and Theory (GMTSAR) *Aug 2026*
Selected from 300+ applicants.

EarthScope Seismology Skill Building Workshop (SSBW) *Jun -- Aug 2026*
70-hour virtual course; Linux & GMT micro-credential.

Geodesy for a Resilient Africa Summer School *Jun -- Jul 2026*
AIMS-Cameroon, Limbe. Built a seismic early-warning web app.

Imperial-AIMS Global Fellows Programme *Jun 2026*
Accra, Ghana. Team lead (PUM-CLID); Best Innovative Accelerator Project.

NASA ARSET: Fundamentals of Remote Sensing -- certificate *Oct 2025*

Ministry of Education, Rwanda: Data Preparation & Analysis Workshop *Sep -- Oct 2024*

ICTP-EAIFR-IUGG Workshop on Computational Geodynamics; 7th ASESMA; Jobberman Finishing School *May -- Jul 2023*

African School of Fundamental Physics and Applications (ASP2022) *Nov -- Dec 2022*
Nelson Mandela University, Gqeberha, South Africa.

African School of Fundamental Physics and Applications (ASP2021) *Jul 2021*
Online.

SELECTED PROJECTS

HeatAlert Accra -- SolarCool CoolTree *Jun 2026*
LSTM-based WBGT heat forecast with CoolShade and CoolBand concepts; dashboard on Hugging Face Spaces. Imperial-AIMS Global Fellows.

TembeaAfrica AI *Mar -- Apr 2026*
Team 5 tool-calling AI travel assistant for Africa (tembeaafrica.github.io). Best Project Award, AIMS-RIC Doctoral Training School, Foundational Methods in Data Science, Kigali.

Seismic Early Warning App *Jun -- Jul 2026*
Seismic data visualisation and alerting (seis-ews-five.vercel.app). Geodesy for a Resilient Africa Summer School.

Object Detection Demo for High Schools *2025*
Real-time item detection for STEM outreach (Google Colab).

AWARDS & HONOURS

Certificate of Appreciation, oral presenter -- 2nd Rwanda Climate Science Symposium *Sep 2026*

Best Innovative Accelerator Project (Team PUM-CLID) -- Imperial-AIMS Global Fellows, Accra *Jun 2026*

Intro Linux & GMT Micro-Credential -- EarthScope / Miami University *Jun 2026*

Best Project Award (TembeaAfrica AI) -- AIMS-RIC Doctoral Training School *Apr 2026*

Best 3-Minute Thesis Award -- ICTP-EAIFR PhD & Master's Competition *Jul 2024*

Mastercard Foundation Scholarship -- MSc at AIMS-Senegal *2021 -- 2022*

SKILLS

Programming: Python, R, C/C++, MATLAB, Fortran, Linux shell, LaTeX
Data science & ML: deep learning, machine learning, LSTM, computer vision, statistical learning
Earth observation: SAR/InSAR, optical EO, GPS geodesy, GMTSAR, GMT, remote sensing
IT & EdTech: Canvas, Moodle, web development, ICDL training, Git, Hugging Face Spaces
Languages: English, Swahili, Luhya (fluent)

MEMBERSHIPS

AXIOM Research Group (AIMS-RIC) · Teachers Service Commission (TSC), Kenya · African School of Fundamental Physics and Applications (ASP) · African Strategy for Fundamental and Applied Physics (ASFAP)